Question 1

A system of gas increases in volume from $0.7\,\mathrm{m}^3$ to $1.8\,\mathrm{m}^3$ at a constant pressure of $52\,\mathrm{kPa}$. Calculate the work in kJ done *on* the gas during this process.

The work done on or by a gas during a process at constant pressure can be calculated using the formula:

$$W = P\Delta V$$

where:

- W is the work,
- P is the constant pressure,
- ΔV is the change in volume.

Given:

- $P = 52$ kPa (1 kPa $= 10^3$ Pa),
- Initial volume $V_i = 0.7\ m^3$,
- Final volume $V_f = 1.8\ m^3$,
- $\Delta V = V_f - V_i$.

Remember our convention: Work done on the system is negative, work done by the system is positive.

$$W = P\Delta V = 52\,\mathrm{kPa} \times 1.1\,\mathrm{m}^3$$

$$W = -57.2\,\mathrm{kJ}$$

The negative sign is to account for our convention.

Question 2

In the given pV-diagram (not to scale, Fig. 1), $p_1 = 104$ kPa, $p_2 = 194$ kPa, $V_1 = 1.2\ m^3$, $V_2 = 2\ m^3$, and $V_3 = 2.9\ m^3$. Calculate the work in kJ done **by** the system during these processes.

The diagram shows a multi-step process:

The total work done on the system, W_{on}, is the sum of the work done in each step.

$$W_{total} = \frac{1}{2}(V_2 - V_1)(p_3 - p_2) + p_2(V_2 - V_1) + \frac{1}{2}(V_3 - V_2)(p_2 - p_1) + p_1(V_3 - V_2) = 607.095 \quad kJ$$

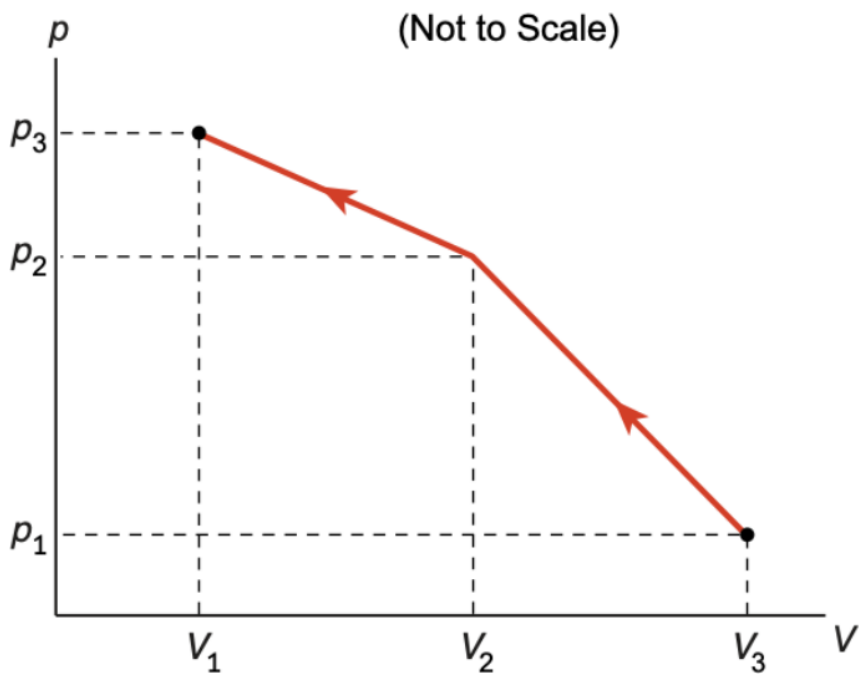

Figure 1: pV-diagram

I did the calculation from left to right (the opposite of what happened in reality), but this gave me the exact sign (work done on the system!). If you follow the natural path (given by the arrows on the pV-diagram), you will end up with a negative work (the gas is being compressed). You need to take it opposite since the work is done on the system.

Question 3

In the given pV-diagram (not to scale, Fig. 1), $p_1 = 114.5$ kPa, $p_2 = 272.2$ kPa, $p_3 = 320.7$ kPa; $V_1 = 0.9\ m^3$, $V_2 = 2.1\ m^3$, and $V_3 = 3.4\ m^3$. Calculate the work in kJ done on the system during these processes.

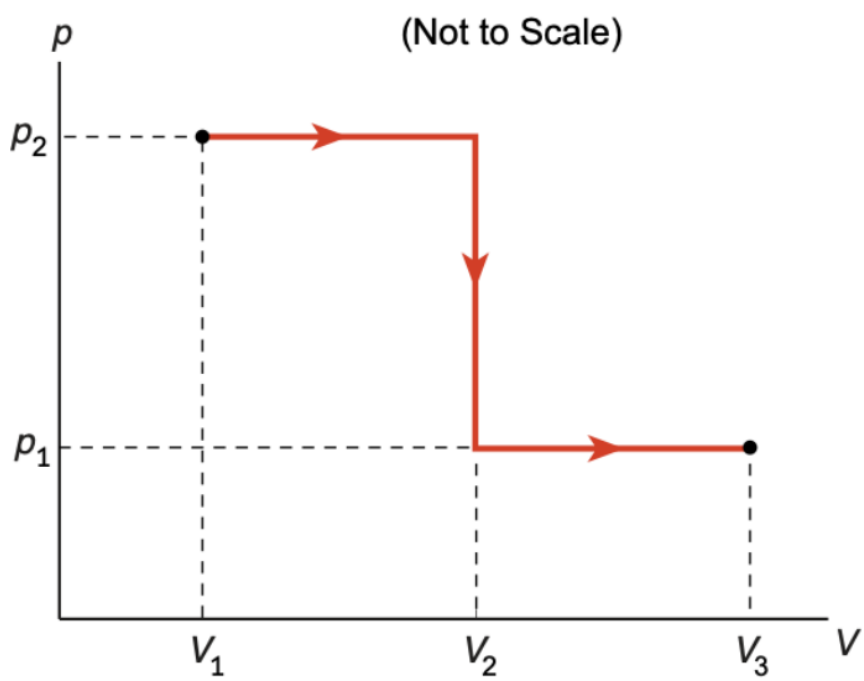

Figure 2: pV-diagram

The diagram shows a multi-step process:

The total work done by the system, W_{on}, is the sum of the work done in each step.

$$W_{total} = p_2(V_2 - V_1) + p_1(V_3 - V_2) = 248.8\ kJ$$

Question 4

Suppose the internal energy of a system of gas changes by 157 J while it does 297 J of work on the environment. Calculate the energy transfer by heat, where it is positive going into the system and negative leaving the system.

The first law of thermodynamics states that the change in the internal energy of a system (ΔU) is equal to the heat added to the system (Q) minus the work done by the system on the surroundings (W):

$$\Delta U = Q - W$$

If the work is done by the system on the environment, it's considered positive in physics, the formula can be rearranged to solve for Q:

$$Q = \Delta U + W$$

Given:

* $\Delta U = 157$ J (change in internal energy),

* $W = 297$ J (work done by the system on the environment).

We calculate Q as follows:

$$Q = 157\,\text{J} + 297 = 454\,\text{J}$$

The positive sign indicates that heat is transferred into the system.

Question 5

An ideal monatomic gas increases in pressure from 134 kPa to 226 kPa at a constant volume of 0.7 m^3. Calculate the energy in kJ transferred by heat during the process.

For an ideal monatomic gas at constant volume, the process is isochoric, so the work done on or by the gas is zero. The first law of thermodynamics, which relates the change in internal energy (ΔU) to the heat added to the system (Q) and the work done by the system (W), simplifies to:

$$\Delta U = Q$$

since $W = 0$ (no work is done during a constant volume process).

The change in internal energy for a monatomic ideal gas can also be calculated using the formula:

$$\Delta U = nC_V \Delta T$$

where:

- n is the number of moles of the gas,

- C_V is the molar heat capacity at constant volume,

- ΔT is the change in temperature.

For a monatomic ideal gas, C_V is $\frac{3}{2}R$, and the ideal gas law ($PV = nRT$) allows us to solve for n and ΔT.

The ideal gas law is:

$$P_1 V = nRT_1$$
$$P_2 V = nRT_2$$

So, for a constant volume process:

$$\frac{P_1}{nR} = T_1$$
$$\frac{P_2}{nR} = T_2$$

The change in temperature (ΔT) can be expressed as:

$$\Delta T = T_2 - T_1 = \frac{P_2 V}{nR} - \frac{P_1 V}{nR} = \frac{V}{nR}(P_2 - P_1)$$

The change in internal energy (ΔU) for a monatomic ideal gas is given by:

$$\Delta U = \frac{3}{2} nR\Delta T$$

Substituting ΔT into the equation for ΔU, we get:

$$\Delta U = \frac{3}{2} nR \left(\frac{V}{nR}(P_2 - P_1) \right) = \frac{3}{2} V(P_2 - P_1)$$

Given:

- $P_1 = 134$ kPa $= 134 \times 10^3$ Pa

- $P_2 = 226$ kPa $= 226 \times 10^3$ Pa

- $V = 0.7 \quad m^3$

We have:

$$\Delta U = \frac{3}{2} V(P_2 - P_1)$$

Converting pressures to Pascals and using the volume in cubic meters, we can plug in the values:

$$\Delta U = \frac{3}{2} \times 0.7 \times (226000 - 134000) = 96.6 \text{ kJ}$$

Question 6

A diatomic ideal gas at constant volume of 1.5 m^3 decreases pressure from 135 kPa to 80 kPa. Calculate the energy transfer by heat in kJ during this process.

In a constant volume process for a diatomic ideal gas, the change in internal energy (ΔU) is directly related to the change in temperature and hence to the change in pressure since V and n are constants. The first law of thermodynamics states that for a system at constant volume, the heat added (Q) is equal to the change in internal energy (ΔU), as no work is done ($W = 0$):

$$Q = \Delta U$$

For a diatomic gas (assuming it behaves ideally and the rotational modes are fully active but vibrational modes are not), the molar heat capacity at constant volume C_V is $\frac{5}{2}R$ due to the contribution of translational and rotational degrees of freedom. The change in internal energy can be calculated by:

$$\Delta U = nC_V \Delta T$$

Since we do not have the number of moles n, we can relate the change in temperature to the change in pressure for a constant volume process using the ideal gas law $PV = nRT$:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

The change in temperature can be related to the change in pressure (at constant volume and constant amount of gas):

$$\frac{P_2 - P_1}{T_1} = \frac{P_2}{T_2} - \frac{P_1}{T_1}$$

Given that the initial and final temperatures are proportional to the initial and final pressures respectively, we can write:

$$\Delta T = \frac{T_2 - T_1}{P_2 - P_1}(P_2 - P_1)$$
$$\Delta T = T_2 - T_1$$

Since $P_1 V = nRT_1$ and $P_2 V = nRT_2$, it follows that:

$$\Delta T = \frac{V}{nR}(P_2 - P_1)$$

Substituting ΔT into the change in internal energy equation, we have:

$$\Delta U = nC_V \frac{V}{nR}(P_2 - P_1)$$

$$\Delta U = \frac{5}{2}V(P_2 - P_1)$$

Given the values:

- $P_1 = 135$ kPa $= 135 \times 10^3$ Pa
- $P_2 = 80$ kPa $= 80 \times 10^3$ Pa
- $V = 1.5m^3$

We can now calculate the change in internal energy (ΔU) and hence the heat added (Q) in kJ:

$$\Delta U = \frac{5}{2} \times 1.5 \times (80 \times 10^3 - 135 \times 10^3)$$

$$\Delta U = \frac{5}{2} \times 1.5 \times (-55 \times 10^3)$$

$$\Delta U = -206.25 \times 10^3 \text{ J}$$

$$\Delta U = -206.25 \text{ kJ}$$

The negative sign indicates that heat is leaving the system.

Question 7

A monatomic ideal gas expands at constant pressure of 135 kPa from $1.5m^3$ to $4m^3$. Calculate the energy in kJ transferred by heat during the process.

For an ideal gas, the heat exchanged is equal to the change in internal energy (ΔU) plus the work done on or by the system (W).

According to the first law of thermodynamic, the heat added to the system (Q) at constant pressure is:

$$Q = \Delta U + W$$

The work done by the gas during an isobaric expansion is given by:

$$W = P\Delta V$$

For a monatomic ideal gas, the change in internal energy (ΔU) can be calculated using the molar heat capacity at constant volume (C_V), which is $\frac{3}{2}R$ per mole for a monatomic ideal gas, where R is the ideal gas constant:

$$\Delta U = nC_V\Delta T$$

Since we do not have the number of moles n, we can relate the change in temperature to the change in pressure for a constant volume process using the ideal gas law $PV = nRT$: Since $PV_1 = nRT_1$ and $PV_2 = nRT_2$, it follows that:

$$\Delta T = P\frac{(V_2 - V_1)}{nR}$$

Substituting ΔT into the change in internal energy equation, we have:

$$\Delta U = nC_V\frac{V}{nR}(V_2 - V_1)$$

$$\Delta U = \frac{3}{2}P(V_2 - V_1)$$

$$Q = \frac{3}{2}P(V_2 - V_1) + P(V_2 - V_1)$$

$$Q = \frac{5}{2}P(V_2 - V_1)$$

$$Q = \frac{5}{2} \times 135 \times (4 - 1.5)$$

$$Q = 843.75\ kJ$$

Question 8

A system of diatomic ideal gas contracts from a volume of 5.2 m^3 to a final volume of 2.2 m^3. If the process proceeds at constant pressure of 190 kPa, calculate the energy in kJ transferred by heat, where positive is energy transferred into the system, and negative is energy transferred out of the system.

For an ideal gas, the heat exchanged is equal to the change in internal energy (ΔU) plus the work done on or by the system (W).

According to the first law of thermodynamic, the heat added to the system (Q) at constant pressure is:

$$Q = \Delta U + W$$

The work done by the gas during an isobaric expansion is given by:

$$W = P\Delta V$$

For a diatomic ideal gas, the change in internal energy (ΔU) can be calculated using the molar heat capacity at constant volume (C_V), which is $\frac{5}{2}R$ per mole for a diatomic ideal gas, where R is the ideal gas constant:

$$\Delta U = nC_V\Delta T$$

Since we do not have the number of moles n, we can relate the change in temperature to the change in pressure for a constant volume process using the ideal gas law $PV = nRT$: Since $PV_1 = nRT_1$ and $PV_2 = nRT_2$, it follows that:

$$\Delta T = P\frac{(V_2 - V_1)}{nR}$$

Substituting ΔT into the change in internal energy equation, we have:

$$\Delta U = nC_V \frac{V}{nR}(V_2 - V_1)$$

$$\Delta U = \frac{5}{2}P(V_2 - V_1)$$

$$Q = \frac{5}{2}P(V_2 - V_1) + P(V_2 - V_1)$$

$$Q = \frac{7}{2}P(V_2 - V_1)$$

$$Q = \frac{7}{2} \times 190 \times (2.2 - 5.2)$$

$$Q = -1995\ kJ$$

Question 9

A monatomic ideal gas undergoes an isothermal compression from a volume of 2.7 m^3 and pressure 103 kPa to a final volume of 1.2 m^3. Calculate the work in kJ done by the gas on its environment.

Since this is an isothermal process, we can calculate the work done using the initial and final pressures and volumes since they are related by $P_1V_1 = P_2V_2$ (Derived from ideal gas equation). We can integrate the work done from the initial to the final volume at a constant temperature, which gives us:

$$W = \int_{V_1}^{V_2} P\, dV$$

For an isothermal process, $P = \frac{nRT}{V}$, so the work done by the gas is:

$$W = \int_{V_1}^{V_2} \frac{nRT}{V}\, dV = nRT \ln\left(\frac{V_2}{V_1}\right)$$

However, since we don't have n and T, we can use the initial state (P_1, V_1) and the ideal gas law to calculate nRT:

$$nRT = P_1V_1$$

Now we can substitute P_1V_1 into our work formula and find the work done by the gas:

$$W = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

Given the initial pressure $P_1 = 103$ kPa (which we convert to Pa by multiplying by 1000), the initial volume $V_1 = 2.7\ m^3$, and the final volume $V_2 = 1.2$ m$^{\mathbf{3}}$, we can calculate the work done:

$$W = 103 \times 10^3 \times 2.7 \ln\left(\frac{1.2}{2.7}\right)$$

We need to calculate this work and remember to convert the result from joules to kilojoules by dividing by 1000.

Using the provided formula and converting the pressure to Pascals, we have:

$$W = 103 \times 10^3 \times 2.7 \times \ln\left(\frac{1.2}{2.7}\right)$$

To calculate the natural logarithm manually or with a calculator, you would do:

$$W = 103 \times 10^3 \times 2.7 \times \ln\left(\frac{1.2}{2.7}\right)$$

$$W = 103 \times 10^3 \times 2.7 \times \ln\left(\frac{4}{9}\right)$$

The natural logarithm of $\frac{4}{9}$ is approximately -0.81. Now we can calculate W:

$$W = 103 \times 10^3 \times 2.7 \times -0.81$$

$$W = 103 \times 10^3 \times -2.187$$

$$W \approx -225.51 \times 10^3 \text{ J}$$

$$W \approx -225.51 \text{ kJ}$$

Thus, the work done by the gas on its environment is approximately -225 kJ (isothermal compression process, the gas has been worked on).

Question 10

A monatomic ideal gas expands at constant temperature from 0.6 m^3 to 2 m^3. If the initial pressure is 169 kPa, calculate the work in kJ done **on** the gas during this process.

For an isothermal expansion of an ideal gas, the work done on the gas can be calculated using the following integral, which derives from the definition of work in a thermodynamic system:

$$W = \int_{V_i}^{V_f} P\, dV$$

Since the process is isothermal, $PV = nRT$ remains constant (Boyle's Law), where n is the number of moles of gas, R is the ideal gas constant, and T is the temperature in Kelvin. We can express pressure P in terms of nRT and V, and substitute into the work integral:

$$W = \int_{V_i}^{V_f} \frac{nRT}{V}\, dV$$

This becomes:

$$W = nRT \ln\left(\frac{V_f}{V_i}\right)$$

However, since $nRT = P_i V_i$ for the initial state, we can rewrite this as:

$$W = P_i V_i \ln\left(\frac{V_f}{V_i}\right)$$

where:

- P_i is the initial pressure,
- V_i is the initial volume,
- V_f is the final volume.

Given that the initial pressure $P_i = 169$ kPa and converting it to Pascals by multiplying by 10^3, the initial volume $V_i = 0.6$ m$^{\mathbf{3}}$, and the final volume $V_f = 2$ m^3, we can calculate the work done on the gas during this expansion:

$$W = 169 \times 10^3 \times 0.6 \ln\left(\frac{2}{0.6}\right)$$

The work done on the gas is positive because the gas is expanding. Now we can calculate W and convert it from joules to kilojoules for the final answer.

The work done on the gas during an isothermal process is given by the equation:

$$W = P_i V_i \ln\left(\frac{V_f}{V_i}\right)$$

Plugging in the given values:

$$W = 169 \times 10^3 \times 0.6 \ln\left(\frac{2}{0.6}\right)$$

The natural logarithm of $\frac{10}{3}$ is approximately 1.204.

$$W = 169 \times 10^3 \times 0.6 \times 1.204$$

$$W \approx 169 \times 0.6 \times 1.204 \times 10^3$$

$$W \approx 122.142 \times 10^3$$

$$W \approx 122.142 \text{ kJ}$$

Therefore, the work done **on** the gas during the isothermal expansion is approximately -122.142 kJ.

Question 11

A diatomic ideal gas is compressed adiabatically from 2.2 m^3 to 1 m^3. If the initial pressure is 90 kPa, calculate the final pressure in kPa.

For an adiabatic process, the relationship between pressure and volume is given by:

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

where:

- P_1 and P_2 are the initial and final pressures,
- V_1 and V_2 are the initial and final volumes,
- γ (gamma) is the heat capacity ratio (also known as the adiabatic index).

For a diatomic ideal gas, γ is typically 1.4. We can rearrange the adiabatic relation to solve for the final pressure P_2:

$$P_2 = P_1 \left(\frac{V_1}{V_2}\right)^\gamma$$

Given:

- $P_1 = 90$ kPa
- $V_1 = 2.2$ m^3
- $V_2 = 1$ m^3
- $\gamma = 1.4$

We can substitute these values to find the final pressure P_2:

$$P_2 = 90 \left(\frac{2.2}{1}\right)^{1.4} = 271.4153\ kPa$$

Question 12

A diatomic ideal gas decreases in pressure adiabatically from an initial pressure of 1.50×10^5 Pa and volume of 1.00 m^3 to a final volume of 2.00 m^3.

(a) Calculate the final pressure in kPa.

(b) Calculate the change in the internal energy in kJ.

For an adiabatic process involving a diatomic ideal gas, the relationship between the initial and final states of pressure and volume is given by:

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

where γ is the heat capacity ratio, which is 1.4 for a diatomic ideal gas.

(a) Calculate the Final Pressure in kPa

Given: - $P_1 = 1.50 \times 10^5$ Pa - $V_1 = 1.00 m^3$ - $V_2 = 2.00 m^3$ - $\gamma = 1.4$

The final pressure P_2 can be found using the adiabatic relationship:

$$P_2 = P_1 \left(\frac{V_1}{V_2}\right)^\gamma$$

Let's calculate P_2 and convert the result to kPa by dividing by 10^3.

$$P_2 = 1.50 \times 10^5 \left(\frac{1.00}{2.00}\right)^{1.4}$$

$$P_2 = 56.8\ kPa$$

(b) Calculate the Change in the Internal Energy in kJ

The change in internal energy (ΔU) for an adiabatic process can be related to work, since $\Delta U = Q - W$, and for an adiabatic process $Q = 0$, so $\Delta U = -W$.

The work done W in an adiabatic process is:

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$W = -90.8\ kJ$$

Question 13

A diatomic ideal gas expands adiabatically from an initial pressure of 179 kPa and volume of 1.00 m^3 to a final volume of 1.9 m^3. Calculate the change in the internal energy in kJ.

To calculate the change in internal energy (ΔU) during an adiabatic expansion for a diatomic ideal gas, we can use the relationship between work (W) and internal energy, knowing that in an adiabatic process, $Q = 0$, so $\Delta U = Q - W = -W$.

The work done in an adiabatic process can be found using the formula:

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

However, to find W, we need to know P_2, the final pressure. This can be calculated using the adiabatic process formula:

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

where $\gamma = \frac{C_p}{C_v}$ is the heat capacity ratio. For a diatomic ideal gas, γ is typically 1.4. Let's first calculate P_2 and then use it to calculate W and consequently ΔU.

Given:

- $P_1 = 179$ kPa
- $V_1 = 1.00$ m^3
- $V_2 = 1.9$ m^3
- $\gamma = 1.4$

$$P_2 = P_1 \left(\frac{V_1}{V_2}\right)^\gamma$$

After calculating P_2, we can find W and ΔU as follows:

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

And since $\Delta U = -W$, we can determine the change in internal energy.

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$W = -101.3283\ kJ$$

Question 14

On a pV-diagram, a monatomic gas process proceeds along the straight line from (3 m^3, 167 kPa) to (5 m^3, 418 kPa). Calculate the energy transferred by heat in kJ.

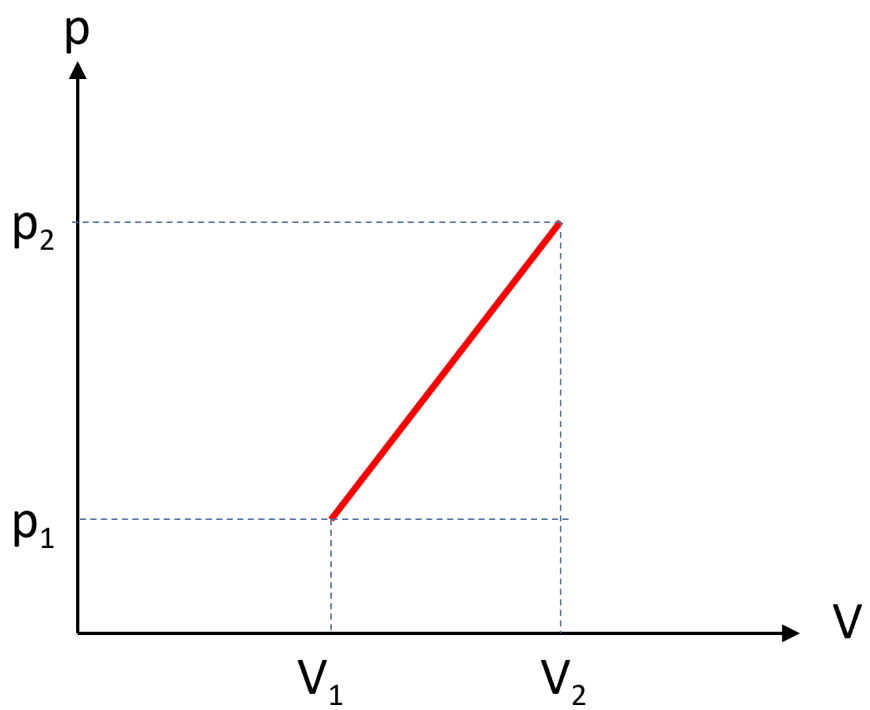

Figure 3: pV-diagram

According to the first law of thermodynamic, the heat added to the system (Q) is:

$$Q = \Delta U + W$$

First we need to account for the change in internal energy.

$$\Delta U = nC_V \Delta T = \frac{3}{2} nR\Delta T$$

Since we do not have the number of moles n, we can relate the change in temperature to the change in pressure-volume using the ideal gas law $PV = nRT$:
Since $P_1V_1 = nRT_1$ and $P_2V_2 = nRT_2$, it follows that:

$$\Delta T = \frac{P_2V_2 - P_1V_1)}{nR}$$

Thus,

$$\Delta U = \frac{3}{2} nR \frac{P_2V_2 - P_1V_1)}{nR} = \frac{3}{2}(P_2V_2 - P_1V_1) = 2.3835 \times 10^3 \ kJ$$

The total work done on the system, W_{on}, is the sum of the work done in each step (3).

$$W_{total} = \frac{1}{2}(V_2 - V_1)(p_2 - p_1) + p_1(V_2 - V_1) = 585 \quad kJ$$

$$Q = \Delta U + W = 2.9685 \times 10^3 \ kJ$$